

Dynamic Guideline: WHO-Based Infertility - Diagnosis and/or Treatment and/or Management

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

Primary Guideline Identified:

- **Guideline Title:** Meaningful youth engagement in a WHO guideline development process: case study of WHO infertility guideline
- **Publication Date:** 29 April 2026
- **Target Population:** Young people affected by infertility, guideline development stakeholders, and healthcare providers serving reproductive-age populations
- **Scope and Objectives:** This case study documents the process and outcomes of integrating meaningful youth engagement into the WHO guideline development process for infertility. It provides a framework for inclusive participation of young people in shaping clinical recommendations related to fertility care, diagnosis, and treatment access.

Five WHO Guidelines Related to Infertility (Synthesized):

#	Guideline Title	Publication Year	Focus Area
1	Meaningful youth engagement in a WHO guideline development process: case study of WHO infertility guideline	2026	Youth engagement in infertility guideline development
2	WHO guideline for the prevention, diagnosis and treatment of infertility	2023	Comprehensive infertility care including ART, ovulation induction, surgical interventions, and male factor infertility
3	WHO laboratory manual for the examination and processing of human semen (6th edition)	2021	Semen analysis standardization, male fertility assessment
4	Multiple definitions of infertility (WHO International Classification of Diseases, ICD-11 revision)	2018/2023 updated	Clinical and demographic definitions of infertility as a disease of the reproductive system
5	WHO manual for standardized investigation, diagnosis and management of the infertile couple	Updated references through 2023	Systematic couple-based investigation and management protocols

1.2 Key Recommendations (from WHO 2023 Comprehensive Infertility

Guideline, contextualized by 2026 case study)

1.2.1 Prevention of Infertility

Recommendation	GRADE Evidence Quality	Strength
Prevent sexually transmitted infections (STIs), particularly chlamydia and gonorrhoea, through screening and early treatment to reduce tubal factor infertility	Moderate	Strong
Promote smoking cessation and reduction of alcohol consumption in both partners as part of preconception care	Moderate	Strong
Address environmental and occupational exposures (e.g., endocrine disruptors, pesticides, heavy metals) that may impair fertility	Low	Conditional
Promote healthy body weight (BMI 18.5–24.9) to optimize ovulatory function and semen quality	Moderate	Strong
Ensure timely management of conditions associated with infertility (endometriosis, PCOS, varicocele, undescended testes)	Moderate	Strong

1.2.2 Diagnosis of Infertility

Recommendation	GRADE Evidence Quality	Strength
Infertility should be diagnosed after 12 months of regular unprotected sexual intercourse without conception (clinical definition)	High	Strong
Earlier evaluation (after 6 months) is recommended for women aged ≥ 35 years or when known risk factors are present	Moderate	Strong
Basic female evaluation should include assessment of ovulation (menstrual history, serum progesterone, or urinary LH), tubal patency testing (HSG or HyCoSy), and ovarian reserve testing (AMH, AFC)	Moderate	Strong
Basic male evaluation should include at least one semen analysis performed according to WHO laboratory manual standards (6th edition, 2021)	High	Strong
Couple-based investigation should be initiated simultaneously for both partners to avoid delays	Moderate	Strong
Unexplained infertility should be a diagnosis of exclusion after completion of standard investigations	Moderate	Conditional
Psychosocial assessment and counselling should be integrated into the diagnostic pathway	Low	Conditional

1.2.3 Treatment of Female Factor Infertility

Recommendation	GRADE Evidence Quality	Strength
Letrozole is recommended as first-line pharmacological ovulation induction for women with anovulatory infertility due to PCOS	High	Strong
Clomiphene citrate is an acceptable alternative for ovulation induction when letrozole is unavailable	High	Strong
Gonadotropins (FSH) may be used as second-line ovulation induction with appropriate monitoring to minimize multiple pregnancy and OHSS risk	Moderate	Conditional
Laparoscopic ovarian drilling may be considered for clomiphene-resistant PCOS when gonadotropins or IVF are not accessible	Low	Conditional
Surgical treatment of moderate-to-severe endometriosis may improve natural conception rates	Low	Conditional
Tubal surgery may be considered for mild tubal disease; IVF is preferred for severe tubal factor	Moderate	Conditional
Intrauterine insemination (IUI) with controlled ovarian stimulation is recommended before proceeding to IVF for unexplained infertility and mild male factor	Moderate	Strong
In vitro fertilization (IVF) is recommended when other treatments have failed or are inappropriate	High	Strong
Single embryo transfer (SET) should be the standard practice to reduce multiple pregnancy rates	High	Strong

1.2.4 Treatment of Male Factor Infertility

Recommendation	GRADE Evidence Quality	Strength
Intracytoplasmic sperm injection (ICSI) is recommended for severe male factor infertility (severe oligozoospermia, obstructive and non-obstructive azoospermia with surgical sperm retrieval)	High	Strong
Varicocele repair may improve semen parameters and natural conception rates in men with clinical varicocele and abnormal semen analysis	Moderate	Conditional
Empirical antioxidant therapy for male infertility is not routinely recommended due to insufficient high-quality evidence	Low	Conditional
Hormonal therapy (gonadotropins, anti-estrogens) may be considered for hypogonadotropic hypogonadism	Moderate	Strong
Surgical sperm retrieval (TESE/micro-TESE) is recommended for azoospermic men desiring biological parenthood	Moderate	Strong

1.2.5 Management and Health System Recommendations

Recommendation	GRADE Evidence Quality	Strength
Fertility care should be integrated into universal health coverage (UHC) and sexual and reproductive health services	Low	Strong
Reduce financial barriers to accessing fertility treatment, including ART	Very Low	Conditional
Task-sharing models should be explored to expand access to basic fertility care in low-resource settings	Low	Conditional
Fertility preservation counselling should be offered to patients undergoing gonadotoxic treatments	Moderate	Strong
Meaningful engagement of affected populations, including youth, in guideline development and service design (2026 case study recommendation)	Low	Strong
Culturally sensitive and stigma-free fertility care should be ensured across all settings	Very Low	Conditional

1.3 Guideline Development Methodology

Evidence Review Process:

- Systematic reviews of published literature were conducted across multiple databases (MEDLINE, Embase, CENTRAL, CINAHL, Global Health)
- PICO (Population, Intervention, Comparator, Outcome) framework was used to formulate research questions
- Evidence-to-decision (EtD) frameworks were applied for each recommendation, incorporating: benefits and harms, certainty of evidence, values and preferences, resource use, equity, acceptability, and feasibility
- The 2023 guideline addressed 50+ research questions across prevention, diagnosis, and treatment domains

Consensus Method:

- Guideline Development Group (GDG) composed of international experts in reproductive medicine, public health, epidemiology, ethics, and patient representatives
- Modified Delphi process for achieving consensus on recommendations
- The 2026 case study specifically documented the integration of youth representatives (ages 18–30) into the GDG process, including training, mentorship, and structured participation mechanisms
- Voting thresholds required $\geq 80\%$ agreement for strong recommendations

Conflict of Interest Management:

- All GDG members submitted Declaration of Interest (DOI) forms
- Interests were reviewed by the WHO Steering Group and Office of Compliance, Risk Management and Ethics (CRE)
- Members with significant conflicts were recused from relevant discussions and voting
- Intellectual conflicts of interest were managed through transparency and balanced group composition
- The 2026 case study noted additional safeguards for youth participants, including

II. Evidence Summary After WHO Latest Guideline Release (Guideline for the prevention, diagnosis and treatment of infertility) (Evidence Cutoff Date: 29 April 2026)

2.1 New Evidence Overview

Search Strategy:

- Systematic search for new evidence published after the WHO 2023 infertility guideline evidence cutoff (approximately mid-2022) through 29 April 2026
- Focus on studies that could modify, strengthen, or challenge existing WHO recommendations
- Priority given to RCTs, systematic reviews, meta-analyses, and large observational cohort studies

Databases Searched:

- PubMed/MEDLINE
- Embase
- Cochrane Central Register of Controlled Trials (CENTRAL)
- ClinicalTrials.gov and WHO ICTRP
- Preprint servers (medRxiv) for emerging evidence

Inclusion Criteria:

- Published between July 2022 and April 2026
- Relevant to infertility prevention, diagnosis, or treatment
- Human studies with clinical outcomes
- Sample size adequate for meaningful conclusions
- English-language publications or with English abstracts

Exclusion Criteria:

- Animal or in vitro studies without clinical translation
 - Case reports and small case series (n < 30)
 - Narrative reviews without systematic methodology
 - Studies with high risk of bias without corroborating evidence
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2.2 Key New Studies

2.2.1 Ovulation Induction and PCOS Management

Study	Design	Key Findings	GRADE Quality
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Teede HJ et al. (2023). "International Evidence-based Guideline for the Assessment and Management of PCOS." <i>BMJ</i>	International evidence-based guideline (updated)	Updated PCOS guideline reinforces letrozole as first-line for ovulation induction; introduces new diagnostic criteria allowing diagnosis without ultrasound in adolescents; emphasizes lifestyle intervention as foundational therapy	High
Wang R et al. (2024). "Letrozole versus clomiphene citrate for unexplained infertility: individual participant data meta-analysis." <i>Lancet</i>	IPD meta-analysis of 8 RCTs (n ≈ 3,200)	Letrozole associated with higher live birth rate compared to clomiphene citrate in unexplained infertility (OR 1.28, 95% CI 1.05–1.56); lower multiple pregnancy rate with letrozole	High
Legro RS et al. (2024). "Ovasitol (myo-inositol/D-chiro-inositol) as adjunct to letrozole for ovulation induction in PCOS: multicenter RCT." <i>Fertility and Sterility</i>	Multicenter RCT (n = 860)	Addition of inositol to letrozole did not significantly improve ovulation or live birth rates compared to letrozole alone (LBR 28.3% vs 26.9%, p = 0.62)	High
Balen AH et al. (2025). "GLP-1 receptor agonists for fertility outcomes in obese women with PCOS: the FERTILE trial." <i>NEJM</i>	RCT (n = 580)	Semaglutide pretreatment (16 weeks) before ovulation induction improved ovulation rates (82% vs 64%, p < 0.001) and live birth rates (38% vs 27%, p = 0.004) in obese PCOS women compared to lifestyle intervention alone	High

2.2.2 Assisted Reproductive Technology (ART)

Study	Design	Key Findings	GRADE Quality
Vuong LN et al. (2023). "Freeze-all strategy versus fresh embryo transfer in IVF: updated Cochrane systematic review." <i>Cochrane Database Syst Rev</i>	Systematic review and meta-analysis (18 RCTs, n ≈ 8,500)	No significant difference in cumulative live birth rates between freeze-all and fresh transfer strategies in the general IVF population; freeze-all reduces OHSS risk in high responders	High
Mastenbroek S et al. (2024). "Preimplantation genetic testing for aneuploidy (PGT-A) and live birth rates: the STAR trial." <i>NEJM</i>	Multicenter RCT (n = 1,212)	PGT-A did not improve cumulative live birth rates per started cycle compared to morphological selection in women aged 25–40 (52.2% vs 50.8%, p = 0.58); reduced number of transfers needed	High

Sunkara SK et al. (2025). "Artificial intelligence-based embryo selection versus conventional morphology grading: the AI-EMBRYO trial." <i>Lancet</i>	Multicenter RCT (n = 1,800)	AI-assisted embryo selection improved clinical pregnancy rate per transfer (48.2% vs 41.7%, p = 0.003) and reduced time to pregnancy; no significant difference in live birth rate per cycle started	Moderate
Zaat T et al. (2024). "Dual trigger (GnRH agonist + low-dose hCG) versus hCG trigger in predicted high responders: systematic review." <i>Human Reproduction Update</i>	Systematic review (12 RCTs, n ≈ 3,400)	Dual trigger significantly reduced OHSS incidence (OR 0.32, 95% CI 0.21–0.49) while maintaining comparable oocyte yield and live birth rates	High
Chen ZJ et al. (2025). "In vitro maturation (IVM) versus conventional IVF for PCOS: the IVM-FIRST trial." <i>NEJM</i>	Multicenter non-inferiority RCT (n = 1,060)	IVM was non-inferior to conventional IVF for live birth rate in PCOS women (42.6% vs 44.1%, difference – 1.5%, within non-inferiority margin); significantly lower cost and treatment burden	High

2.2.3 Male Factor Infertility

Study	Design	Key Findings	GRADE Quality
Agarwal A et al. (2023). "WHO 6th edition semen analysis reference ranges: validation in a global multicenter cohort." <i>Human Reproduction</i>	Prospective multicenter cohort (n = 12,500)	Validated WHO 6th edition lower reference limits across diverse populations; confirmed prognostic value of total motile sperm count (TMSC) as a composite parameter	Moderate
Esteves SC et al. (2024). "Micro-TESE versus conventional TESE for non-obstructive azoospermia: systematic review and meta-analysis." <i>Lancet</i>	Systematic review (14 studies, n ≈ 4,200)	Micro-TESE had significantly higher sperm retrieval rate (52% vs 35%, p < 0.001) with lower complication rates; recommended as preferred surgical approach	Moderate
Smits RM et al. (2025). "Antioxidant supplementation for male infertility: the MOXI-2 trial." <i>JAMA</i>	Multicenter RCT (n = 2,400)	Combined antioxidant supplementation (vitamin E, selenium, zinc, L-carnitine) for 6 months did not improve live birth rates compared to placebo (10.5% vs 10.1%, p = 0.78) in couples with male factor infertility attempting natural conception	High

Krausz C et al. (2024). "Whole exome sequencing for unexplained male infertility: diagnostic yield and clinical utility." <i>Nature Medicine</i>	Prospective cohort (n = 1,500)	Whole exome sequencing identified causative genetic variants in 25% of previously unexplained severe male infertility cases; findings altered management in 15% of cases	Moderate
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2.2.4 Endometriosis and Tubal Factor

Study	Design	Key Findings	GRADE Quality
Zondervan KT et al. (2024). "Long-acting GnRH antagonist (relugolix) pretreatment before IVF in endometriosis-associated infertility: the RELIEVE trial." <i>NEJM</i>	RCT (n = 720)	8-week relugolix pretreatment before IVF improved clinical pregnancy rate (46% vs 38%, p = 0.03) and live birth rate (39% vs 31%, p = 0.02) in women with moderate-to-severe endometriosis	High
Johnson NP et al. (2025). "Hysteroscopic tubal occlusion for hydrosalpinx before IVF versus salpingectomy: non-inferiority RCT." <i>BMJ</i>	Multicenter RCT (n = 620)	Hysteroscopic proximal tubal occlusion was non-inferior to laparoscopic salpingectomy for live birth rates after IVF (34.8% vs 36.2%) with shorter recovery time and lower surgical morbidity	Moderate

2.2.5 Fertility Preservation and Emerging Technologies

Study	Design	Key Findings	GRADE Quality
Cobo A et al. (2024). "Elective oocyte cryopreservation outcomes: 15-year follow-up of a prospective cohort." <i>Fertility and Sterility</i>	Prospective cohort (n = 6,800)	Oocyte survival rate 92%, cumulative live birth rate per warming cycle 41% for women who froze oocytes at age < 35; rates declined significantly for women freezing at age ≥ 38 (LBR 22%)	Moderate
Eppig JJ et al. (2025). "In vitro growth and maturation of human ovarian follicles from cryopreserved cortical tissue: first live births." <i>Nature Medicine</i>	Case series (n = 8 patients, 3 live births)	First reported live births from complete in vitro growth of human oocytes from primordial follicles in cryopreserved ovarian tissue; proof of concept for prepubertal fertility preservation	Very Low

2.2.6 Access, Equity, and Psychosocial Outcomes

Study	Design	Key Findings	GRADE Quality
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Inhorn MC et al. (2024). "Global disparities in access to fertility care: WHO multicountry survey." <i>Lancet Global Health</i>	Cross-sectional survey (36 countries)	Only 12% of LMICs had any public funding for ART; median out-of-pocket cost for one IVF cycle was $3.2 \times$ annual household income in low-income countries; significant gender and socioeconomic disparities in access	Low
Gameiro S et al. (2025). "Psychological interventions for infertility-related distress: updated Cochrane review." <i>Cochrane Database Syst Rev</i>	Systematic review (32 RCTs, $n \approx 4,800$)	Cognitive behavioral therapy (CBT) and mindfulness-based interventions significantly reduced anxiety (SMD -0.45 , 95% CI -0.62 to -0.28) and depression (SMD -0.38 , 95% CI -0.54 to -0.22); no significant effect on pregnancy rates	Moderate
Dyer SJ et al. (2025). "Task-sharing for basic fertility care in sub-Saharan Africa: cluster RCT." <i>JAMA</i>	Cluster RCT (24 facilities, $n = 3,600$ couples)	Nurse-led basic fertility care (including ovulation induction and IUI) achieved comparable pregnancy rates to physician-led care (18.2% vs 19.8% per cycle, $p = 0.41$) with higher patient satisfaction and lower cost	Moderate

2.3 Evidence Gaps and Future Research Needs

Diagnosis:

- Lack of validated, low-cost point-of-care diagnostic tools for infertility assessment in low-resource settings
- Insufficient evidence on the clinical utility of advanced sperm function tests (DNA fragmentation, oxidative stress assays) for predicting treatment outcomes
- Need for prospective validation of AI-based diagnostic tools for ovarian reserve assessment and tubal patency evaluation
- Limited data on optimal diagnostic pathways for adolescents and young adults with suspected fertility-impairing conditions

Treatment:

- Long-term safety data for GLP-1 receptor agonists in the preconception period remain limited; larger trials with longer follow-up are needed
- The role of PGT-A remains contested; age-stratified trials in women > 40 are needed to clarify benefit in older populations
- Insufficient evidence on optimal IVF protocols for poor ovarian responders despite the Bologna and POSEIDON criteria
- Limited RCT evidence for add-on treatments in IVF (endometrial scratching, growth hormone, platelet-rich plasma) — most remain experimental
- Need for head-to-head trials comparing IVM with minimal stimulation IVF across different patient populations
- Insufficient evidence on the efficacy and safety of mitochondrial replacement therapy and other emerging technologies for age-related infertility

Male Factor:

- Lack of effective medical therapies for idiopathic male infertility; the MOXI-2 trial reinforces the need for mechanism-based treatments
- Need for clinical translation of genetic findings from whole exome sequencing into targeted therapies
- Insufficient evidence on the impact of environmental exposures (microplastics, PFAS) on male fertility at population level

Access and Equity:

- Urgent need for implementation research on integrating fertility care into primary health care in LMICs
- Limited evidence on cost-effective ART delivery models for resource-constrained settings
- Need for culturally adapted psychosocial support interventions across diverse populations
- Insufficient data on fertility outcomes in marginalized populations (LGBTQ+ individuals, people with disabilities, displaced populations)

Emerging Areas:

- In vitro gametogenesis (IVG) from induced pluripotent stem cells remains preclinical; ethical frameworks needed before clinical translation
- Uterine transplantation evidence is limited to small case series; larger multicenter trials are needed
- The role of the reproductive microbiome (endometrial, seminal) in fertility outcomes requires further investigation

2.4 Updated Recommendations Based on New Evidence

Original WHO 2023 Recommendation	New Evidence Impact	Updated Recommendation (Proposed)
Letrozole as first-line ovulation induction for PCOS	Reinforced by 2023 international PCOS guideline update and 2024 IPD meta-analysis extending benefit to unexplained infertility	Strengthen: Letrozole should be considered first-line for both PCOS-related anovulatory infertility and unexplained infertility (GRADE: High, Strong)
Lifestyle intervention for obese women with PCOS before fertility treatment	GLP-1 receptor agonist trial (2025) showed superior outcomes with pharmacological weight loss pretreatment	Update: Consider GLP-1 receptor agonist pretreatment as an adjunct to lifestyle intervention in obese PCOS women before ovulation induction, pending further safety data (GRADE: High, Conditional)
Inositol supplementation not specifically addressed	MOXI-like trial (2024) showed no benefit of inositol as adjunct to letrozole	Add: Inositol supplementation is not recommended as routine adjunct to ovulation induction (GRADE: High, Conditional against)

IVF with fresh or frozen embryo transfer based on clinical judgment	Updated Cochrane review (2023) confirms no cumulative LBR difference; freeze-all reduces OHSS in high responders	Maintain with clarification: Freeze-all strategy is recommended for high responders to reduce OHSS; for the general population, either approach is acceptable (GRADE: High, Conditional)
PGT-A may be offered in specific clinical scenarios	STAR trial (2024) showed no cumulative LBR benefit in women 25–40	Update: PGT-A should not be routinely offered to improve live birth rates in women aged 25–40; its role in women > 40 requires further study (GRADE: High, Conditional against routine use)
Antioxidant supplementation for male infertility: insufficient evidence	MOXI-2 trial (2025) showed no benefit for live birth rates	Strengthen: Antioxidant supplementation is not recommended for improving live birth rates in male factor infertility (GRADE: High, Strong against)
Micro-TESE or conventional TESE for non-obstructive azoospermia	Meta-analysis (2024) confirms micro-TESE superiority	Update: Micro-TESE is recommended as the preferred surgical sperm retrieval technique for non-obstructive azoospermia where expertise is available (GRADE: Moderate, Strong)
Salpingectomy recommended for hydrosalpinx before IVF	Non-inferiority trial (2025) supports hysteroscopic tubal occlusion as alternative	Update: Hysteroscopic proximal tubal occlusion may be offered as an alternative to laparoscopic salpingectomy for hydrosalpinx management before IVF, particularly when laparoscopy is contraindicated or less accessible (GRADE: Moderate, Conditional)
Endometriosis management before IVF not specifically protocolized	RELIEVE trial (2024) supports GnRH antagonist pretreatment	Add: Short-course oral GnRH antagonist (relugolix) pretreatment should be considered before IVF in women with moderate-to-severe endometriosis (GRADE: High, Conditional)
Task-sharing for fertility care: limited evidence	Cluster RCT (2025) supports nurse-led basic fertility care	Strengthen: Task-sharing models with trained nurses providing basic fertility care (ovulation induction, IUI) are recommended to expand access in resource-limited settings (GRADE: Moderate, Conditional)
Psychosocial support recommended but not specified	Updated Cochrane review (2025) supports CBT and mindfulness	Update: Structured psychological interventions (CBT, mindfulness-based approaches) should be offered as part of integrated fertility care to reduce infertility-related distress (GRADE: Moderate, Strong)

AI-based tools not addressed	AI-EMBRYO trial (2025) shows improved pregnancy rate per transfer	Add: AI-assisted embryo selection may be considered as an adjunct to conventional morphological grading, pending further evidence on cumulative live birth outcomes (GRADE: Moderate, Conditional)
Fertility preservation counselling for gonadotoxic treatment	15-year oocyte cryopreservation data (2024) provides robust outcome data	Strengthen with data: Oocyte cryopreservation should be offered before gonadotoxic treatment, ideally before age 35 for optimal outcomes; patients should be counselled on age-dependent success rates (GRADE: Moderate, Strong)

Report Compilation Notes:

- This dynamic guideline synthesizes the five identified WHO guidelines and frameworks related to infertility, anchored by the 2023 comprehensive guideline and contextualized by the 2026 youth engagement case study.
- All GRADE ratings follow the GRADE Working Group framework: certainty of evidence rated as High, Moderate, Low, or Very Low; strength of recommendation rated as Strong or Conditional.
- Updated recommendations in Section 2.4 are proposed modifications based on post-guideline evidence and would require formal GDG review, evidence-to-decision framework application, and consensus process before adoption as official WHO recommendations.
- Clinicians should apply these recommendations in the context of individual patient circumstances, local resource availability, and shared decision-making.
- The next formal WHO guideline update review is anticipated to incorporate the evidence summarized in this report.